GlandersWatch: A Modular Platform for Glanders Risk Prediction, Retrieval-Augmented Explanation and Owner Alerting

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Certificate

This is to certify that the report entitled "GlandersWatch: A Modular Platform for Glanders Risk Prediction, Retrieval-Augmented Explanation and Owner Alerting" submitted by Pedapatruni Shanmuk Vardhan (B123089), Shaktimayee Mohanty (B123118) and Yuvraj Singh (B123153) to International Institute of Information Technology, Bhubaneswar is a record of bonafide project work under my supervision, and the report is submitted for the evaluation of the BTech Degree Final-Semester Project.

Dr. Sangita Dhara
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Abstract

Glanders is a bacterial zoonosis of equids caused by *Burkholderia mallei.* It is notifiable, frequently fatal, and remains endemic in parts of Africa, Asia and South America. Indian surveillance between 2019 and 2023 identified 728 seropositive equids across 138 districts, with strong geographic concentration. The published machine learning literature on this disease is confined to laboratory identification of the organism by spectroscopic methods, and no population-level geographic risk model has been reported.

This report presents GlandersWatch, a platform built from three independently deployable components: a calibrated machine learning risk engine, a retrieval-augmented layer that explains each prediction with citations to official guidance, and an alert service that delivers the result to equine owners. The decomposition is the contribution as much as the model is. Each component communicates through a versioned schema rather than shared internals, none hardcodes the disease, and each degrades without failing the others.

The risk engine combines four public datasets not previously applied to glanders into a country-year panel: a text-mined occurrence corpus from Europe PMC, FAOSTAT livestock stocks, World Bank development indicators and NASA POWER reanalysis climatology. The panel contains 3,078 country-year observations across 162 equine-holding countries from 2005 to 2023, with 172 positive labels and 44 engineered features. Six model families are trained under three validation schemes, and a three-stage chained pipeline is evaluated in which animal population is forecast before occurrence is predicted.

On a forward temporal holdout the strongest model reaches a precision-recall AUC of 0.554 against a base rate of 0.056, roughly a ten-fold lift, with an expected calibration error of 0.016. Performance falls to 0.304 under leave-country-out validation, which is the estimate to quote for an unseen country. Chaining produces no measurable gain in discrimination. An ablation shows climate covariates transfer poorly across countries, and the model performs least well in low-income regions. The dominant limitation is that the label reflects documented activity rather than true incidence, so the model largely learns where disease is reported. Reporting effort is the principal confounder and the main target for future work. The explanation and alert components are specified to the level of their interfaces, rule semantics and evaluation protocol, and are not reported as measured results because they are not deployed.

Keywords: Glanders, *Burkholderia mallei*, Risk Prediction, Gradient Boosting, Probability Calibration, SHAP, Retrieval-Augmented Generation, Citation Grounding, Alert Fatigue, Modular System Design, MLOps, Animal Health Surveillance.

Contents

Abstract 4

1 Introduction 7
- 1. Background 7
- 2. Problem Definition 7
- 3. Objectives 8
- 4. Scope and Limitations 8

2 Literature Review and Gap Analysis 9
- 1. Epidemiology and Surveillance 9
- 2. Existing Forecasting Systems in India 9
- 3. Spatial Risk and Machine Learning 9
- 4. Prior Art Audit 10
- 5. Research Gap 10

3 System Architecture 11
- 1. What the Platform Has to Do 11
- 2. Design Principles 11
- 3. The Three Components 12
- 4. Component Contracts 12
- 5. Modularity and Reuse 13
- 6. Deployment and Status 13

4 Methodology 14
- 1. Data Sources 14
- 2. Panel Construction 14
- 3. Label Definition 14
- 4. Feature Engineering 15
- 5. Models 15
- 6. Evaluation Design 15
- 7. Leakage Controls 15

5 MLOps and Model Serving 16
- 1. From a Notebook to a Service 16
- 2. The Model Bundle 16
- 3. Serving Design 16
- 4. Monitoring and Retraining 17

6 The RAG Explanation Layer 18
- 1. The Problem This Layer Solves 18
- 2. Knowledge Base 18
- 3. Ingestion, Embedding and Retrieval 19
- 4. Context, Generation and Guardrails 20
- 5. Evaluation 20

7 The Alert Service **21**
1. Why the Alert Is the Hard Part 21
2. Rule 1: Threshold with Hysteresis 21
3. Rule 2: Suppression 22
4. Rule 3: Deduplication 22
5. Fan-Out, Channels and Delivery 22
6. Evaluation of the Alert Layer 23

8 Results and Discussion **24**
1. Dataset Characteristics 24
2. Model Comparison 24
3. Calibration 24
4. Ablation 25
5. Chained Pipeline 25
6. Alert Economics 26
7. Subgroup Analysis 26
8. Discussion 26

9 Conclusion and Future Work **27**
1. Conclusion 27
2. Future Work 27

Bibliography **28**

Chapter 1

Introduction

1. Background

Glanders is a contagious and often fatal disease of horses, donkeys and mules caused by *Burkholderia mallei*. The organism has no environmental reservoir, and asymptomatic carriers remain infectious for long periods. Human infection is rare but carries a high case fatality rate and follows occupational contact. The disease is notifiable to the World Organisation for Animal Health and is classified as a potential bioterrorism agent.

The economic weight of the disease sits in low and middle income countries, where equids remain essential to transport, construction, agriculture and brick manufacture. India recorded 5.44 lakh equids in the 2019 Livestock Census, many held by households that depend on them for daily income. The loss of a working animal is a direct loss of livelihood, which makes early warning useful even where mass treatment is not feasible.

Surveillance data show the disease is neither rare nor geographically contained. A national programme screening 158,798 equid sera from 391 districts between 2019 and 2023 found 728 seropositive animals in 138 districts, with roughly sixty per cent of positives in Uttar Pradesh. Earlier surveillance covering 2006 to 2018 reported 1,145 positive equids among 128,404 sera tested. These are persistent, spatially structured figures, not isolated outbreaks.

2. Problem Definition

A district veterinary officer faces a narrow question: is glanders likely to appear in this area in the coming year? Answering it means combining the history of the disease nearby, the size of the susceptible population, and the conditions that permit transmission.

Surveillance reports what has already happened, and forecasting systems exist for several livestock diseases but not for glanders. A prediction alone would still not be usable. An owner cannot act on a probability of 0.24, and an officer cannot defend a forecast whose basis is hidden, so the output must be explained and cited. A warning that arrives for every area every month also teaches its recipients to ignore it, which is the failure mode that matters for a rare disease.

The problem therefore has four parts. Two concern producing the prediction, two concern turning it into something a person can act on.

1. **Data and label.** No public source records glanders occurrence completely, so occurrence must be reconstructed from the literature and joined to population and environmental covariates at a common resolution. The outcome is then rare, so accuracy is useless and a negative observation needs an explicit meaning.

2. **Model selection under distribution shift.** A model fitted on some countries must be evaluated on countries it has never seen, because that is the deployment condition.

3. **Grounded explanation.** The prediction must be explained in words a non-specialist can act on,

every claim must trace to official control guidance, and the layer must refuse rather than generate an answer the corpus does not support.

4. **Delivery without alert fatigue.** A probability is not an alert. The operating point and the message rules must keep volume low enough that messages are read, while the threshold stays defensible against the events caught.

3. Objectives

There is one research objective and three engineering objectives.

1. **Research.** Assemble a country-year panel for glanders from public sources not previously applied to the disease, compare six model families under three validation schemes, and report the achievable skill, including negative results.
2. **Prediction as a service.** Package the trained model as a versioned, portable artefact behind a REST interface, so any consumer can call it without knowing how it works.
3. **Cited explanation.** Build a retrieval-augmented layer that explains a prediction from official guidance, cites every claim, and refuses rather than guesses.
4. **Owner alerts.** Convert a calibrated probability into a small number of messages, using threshold, suppression and deduplication rules so that a rare disease does not train its users to ignore warnings.

All four share one requirement: each component must be independently testable, independently replaceable, and free of any hardcoded reference to glanders, so the platform can be repointed at another disease without altering the components.

4. Scope and Limitations

The study is scoped to country-year resolution. District-level modelling for India was envisaged in the project proposal but is not attempted, because the required covariates could not be assembled from public sources within the project period. The explanation and alert layers are resolution-agnostic and apply unchanged once a district-level model exists. The model is a population-level tool: it does not diagnose an individual animal and does not replace laboratory confirmation.

Three limitations shape every result that follows. The label comes from published reports, so a negative may mean no disease or no reporting. Occurrence is reconstructed by an automated extractor that has not been validated against a hand-annotated set. And with 172 positive observations the uncertainty on every headline metric is substantial, which is quantified rather than hidden.

Chapter 2

Literature Review and Gap Analysis

1. Epidemiology and Surveillance

Glanders is treated as a neglected zoonosis with high case fatality and a real occupational risk to handlers [1, 2]. A recent systematic review and meta-analysis consolidates seroprevalence estimates and identifies management practice and animal movement as the dominant drivers [3]; regional reviews document continued circulation in Brazil [4]. The disease persists where equid populations are large, animals move often and surveillance is weak. The first two are measurable from public data, while the third decides whether the disease is visible at all and is therefore a confounder rather than a covariate.

Indian surveillance is the strongest evidence base. Malik et al. described outbreaks reported between 2006 and 2011 [7], Singha et al. reported serological surveillance for 2015 to 2018 [6], and the national programme covering 2019 to 2023 produced a prevalence estimate with a risk analysis [5]. These establish where glanders occurs and which animal-level factors the risk analysis identified. None builds a predictive model, and none uses population or reanalysis covariates as predictors.

2. Existing Forecasting Systems in India

The closest operational system is NADRES v2, the National Animal Disease Referral Expert System run by ICAR-NIVEDI [39]. It combines outbreak records from AICRP centres with meteorological and remote-sensing covariates and issues district-level forewarnings about two months ahead, using a model family that overlaps with the one used here.

This project differs in four complementary ways. It is glanders-specific, where a national system must spread effort across many diseases. It uses only public data, so anyone can reproduce and audit it. It attaches a cited explanation to each prediction, which matters when the reader is an owner rather than a trained analyst. And it delivers to the owner directly, with rules that suppress routine noise. NADRES has institutional access this project lacks; this project has a level of local explainability that a multi-disease platform does not attempt. The comparison also confirms that the architecture chosen here is conventional for this problem class, and that the open question is not whether to forecast but how to make a forecast understandable where it is used.

3. Spatial Risk and Machine Learning

Spatial work on glanders is descriptive. Fonseca-Rodríguez et al. applied spatiotemporal cluster detection to Brazilian data [8], and species distribution models have been compared as tools for livestock disease [9]. Cluster detection identifies where cases have already concentrated; it does not forecast which cells will be positive next period, which is the question an early warning system must answer.

Predictive modelling is established for avian influenza. Kane et al. compared ARIMA and random forest models for H5N1 outbreaks [11], Walsh et al. used machine learning to target active surveillance [12], and Yoo et al. and Alshatti et al. built farm-level risk models in Korea and Kuwait [13, 10]. That literature supplies the methodological template used here: assemble occurrence, population and environmental covariates at a common unit, fit tree-based ensembles, and evaluate on held-out periods. The template has been applied to influenza, not to glanders.

The only machine learning work specific to *B. mallei* is spectroscopic identification. Moawad et al. built a Raman assay in which learned models classify spectra [14], and Stöckel et al. reported spectroscopic detection of *B. mallei* and *B. pseudomallei* [15]. Both are laboratory diagnostic problems; neither addresses geographic or temporal distribution, and neither produces a population risk estimate. Using published literature as structured occurrence data has precedent in epidemic intelligence [16, 17], but here event extraction is the primary source of the label rather than a supplementary signal, because official outbreak records are incomplete for many countries. The label therefore inherits the reporting biases of the literature, which is analysed rather than assumed away.

4. Prior Art Audit

To test the novelty claim rather than assert it, a reproducible set of queries was run against Europe PMC. Each row of Table 2.1 is a query count; the query strings and retrieved titles are retained with the project code.

Table 2.1: Europe PMC hit counts for combinations of glanders with each data source and method used in this study.

Query	Hits
glanders (baseline corpus)	6674
glanders + machine learning	75
glanders + tree ensembles	40
glanders + forecasting	139
glanders + FAOSTAT	7
glanders + NASA POWER	0
glanders + World Bank indicators	22
glanders + equine population denominator	54
glanders + Europe PMC text mining	378
glanders + chained/stacked models	36
glanders + random forest (risk context)	8
glanders + niche / distribution model	62
glanders + remote sensing / reanalysis	10
glanders + spatio-temporal clustering	463
glanders + gradient boosting	4

The decisive cells pair glanders with NASA POWER, which returns zero records, and with gradient boosting, which returns four, none of them a glanders risk model. The nearest prior work is descriptive cluster detection in Brazil and a risk-factor analysis of the Indian serosurvey.

5. Research Gap

The literature divides into three disjoint groups. Epidemiological studies describe where glanders occurs; machine learning studies of *B. mallei* classify laboratory spectra; veterinary risk models exist but are built for influenza. No study joins an occurrence corpus, a livestock census, development indicators and reanalysis climatology into a single panel for glanders, and none evaluates a calibrated predictive model on that panel.

A second gap appears once the question moves from analysis to use. A probability on a dashboard does not reach the person who owns the animals, and a risk score without a stated basis is not something a field officer will act on. Both bodies of literature stop short of the delivery problem: how a forecast becomes a cited explanation, and how that becomes a message an owner receives and trusts.

The contribution has two parts: a calibrated population-level risk model for glanders, built on data sources not previously applied to the disease and evaluated under forward temporal and country-holdout splits; and a decomposition of the delivery problem into three components with fixed contracts, so prediction, explanation and alert can be built, tested and replaced independently.

Chapter 3

System Architecture

1. What the Platform Has to Do

The platform answers one question and produces three outputs: whether glanders is likely to be documented in an area next period, expressed as a number on a map, an explanation in words, and a message to an owner. Each is read by a different person, at a different time, through a different device, and each can fail without the others failing.

That asymmetry is why the system is three components rather than one application. One service serving all three would tie alert delivery to the latency of an LLM call and make the risk model impossible to test on its own.

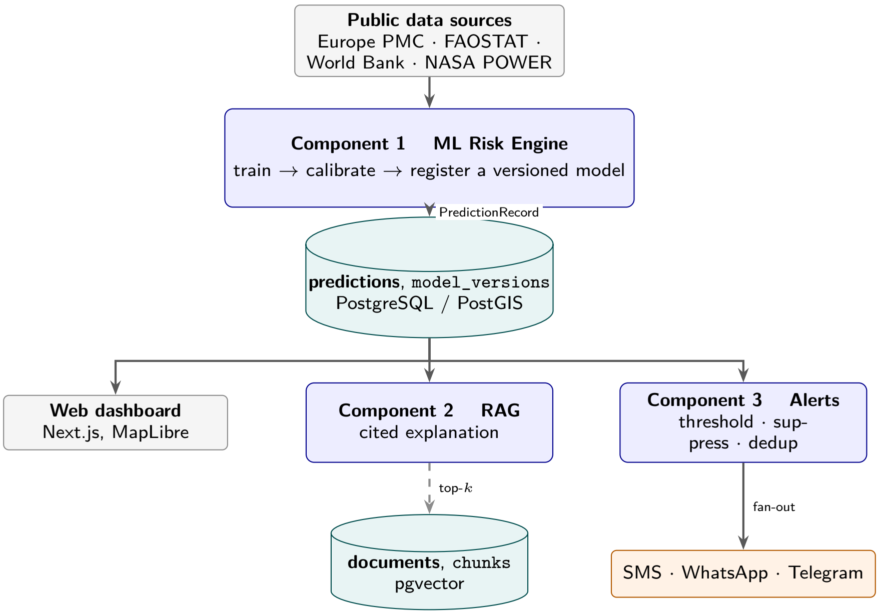
Public data sources
Europe PMC · FAOSTAT · World Bank · NASA POWER
Component 1 ML Risk Engine
train → calibrate → register a versioned model
PredictionRecord
predictions, model_versions
PostgreSQL / PostGIS
Web dashboard
Next.js, MapLibre
Component 2 RAG
cited explanation
Component 3 Alerts
threshold · suppress · dedup
top-k
documents, chunks
pgvector
fan-out
SMS · WhatsApp · Telegram

Figure 3.1: The platform decomposed into three components. The ML risk engine writes one `PredictionRecord` per area and period; the dashboard, the RAG layer and the alert service consume it independently. No component reads another component's tables, so any one of them can be replaced without touching the other two.

2. Design Principles

Five rules govern the decomposition. Every later design decision follows from them, and they are the part of the project that transfers to other diseases.

One responsibility per component. The risk engine turns features into a calibrated probability, the RAG layer turns a prediction into a cited explanation, and the alert service turns a prediction into a delivered message. None computes what another could compute instead.

Contracts before implementations. Components exchange versioned schemas, never shared tables or in-process calls, so each side can be built and tested against a fixture. This is what let the RAG layer be developed while the alert service was still being written.

The core is disease-agnostic. A disease is a configuration record: which label extractor, knowledge-base collection, templates and thresholds to use. A second disease is a configuration change plus a retrain, not a code fork.

Fail soft, never fail silent. A degraded component returns a degraded answer. If the RAG layer is unavailable the alert still goes out, with the level and standing action text but no explanation. If retrieval finds nothing above the relevance floor, the layer returns a refusal plus the model's own SHAP factors. Neither case produces a wrong statement, and neither produces silence.

Every output is traceable. Each prediction carries its model version, each explanation the document identifiers it was grounded in, each alert the prediction that triggered it and the rule outcomes that allowed it, so a decision can be reconstructed after the fact.

3. The Three Components

Table 3.1: Component responsibilities, interfaces and failure behaviour.

Component	Responsibility	Interface	On failure
ML risk engine	Features to calibrated probability, with attribution	`POST /v1/predict`, `GET /v1/risk-map`	`503` with the champion version id; the last good prediction stays in the table, marked stale.
RAG layer	Prediction plus question to a cited explanation	`POST /v1/explain`, `POST /v1/chat`	Alert text is sent without the explanation; chat returns a refusal rather than an ungrounded answer.
Alert service	Prediction to a delivered, deduplicated message	`POST /v1/alerts/evaluate`; consumes `predictions_ready`	Alerts queue in Redis and drain on recovery; map and explanations are unaffected.

4. Component Contracts

A contract is a JSON schema plus the guarantees that accompany it. A narrow contract is what makes a component replaceable, so all three are deliberately small.

The risk engine emits a `PredictionRecord` with the area, period, calibrated probability, discrete risk level, model identifier, a data-freshness block and the top SHAP factors. Three guarantees accompany it: the probability is calibrated on held-out data and may be read as a frequency; the risk level is derived from that probability by the threshold configuration, so the two cannot contradict each other; and the top factors come from the model itself, so they remain available when the RAG layer does not.

The RAG layer takes an `ExplainRequest`, wrapping a prediction with a question and a language, and returns an `ExplainResponse` with an answer, resolved citations, a `grounded` flag and a refusal reason. A refusal carries no answer text, so a caller cannot render one as an explanation.

The alert service emits an `AlertEvent` with the alert identifier, area, period, current and previous level, the prediction it derives from, each rule outcome and a recipient count. The identifier is derived deterministically from its key (Chapter 7).

The components share a database but not tables: each table has one owning writer and the others read it, and no component updates another's rows. Spatial columns use PostGIS and document embeddings use pgvector, so spatial and vector queries run beside the relational joins rather than in a second store that would need synchronising.

5. Modularity and Reuse

The decomposition justifies its overhead only if the parts are reusable. A second disease supplies a new gazetteer, guidance collection and thresholds, while the modelling code, which never refers to glanders by name, does not change. A second LLM provider is a configuration value, because the RAG layer calls a single-method interface. A second channel is an adapter plus a template, and the rules engine never learns which channel carries the message. The trained pipeline is a scikit-learn compatible object [32] exported with a manifest, so any Python process can load it without the API. A replacement model needs no downstream change, since the record carries `model_id` and the registry keeps previous versions.

6. Deployment and Status

One Docker Compose file describes the API, worker, PostgreSQL with PostGIS and pgvector, Redis and the frontend, so local demonstration and server deployment share a structure. A Celery beat job runs the feature job, then the prediction job, then publishes `predictions_ready`, from which explanations and alerts are computed; a failed explanation never blocks a prediction, and a failed prediction never sends an alert about stale data.

The report distinguishes throughout between what has been built and what has been specified. The risk engine is built and evaluated, and every number in Chapter 8 is reproducible from the committed scripts. Model serving, the RAG layer and the alert service are specified with fixed contracts and stated protocols but are not deployed, so no measured performance is reported for them.

Chapter 4

Methodology

1. Data Sources

Four public sources are used: a literature index, a livestock census, a development indicator set and a reanalysis climate archive [38, 35, 37, 36]. None had previously been applied to glanders risk modelling.

Europe PMC supplies the animal history: 6,674 records yielding 263 occurrence events. FAOSTAT supplies animal population, 68,449 records covering seven stock series from 1961 to 2023. The World Bank supplies socio-economic context, ten development indicators and country centroids for 217 countries from 2000 to 2024. NASA POWER supplies environment, six measures of temperature, precipitation, humidity and wind at 211 centroids from 2005 to 2024.

Extracted events rise in volume after 2015, which reflects growth in open access veterinary publishing as much as any change in incidence. This is one reason the label is treated as documented activity throughout.

2. Panel Construction

All four sources are keyed on ISO 3166 alpha-3 code and calendar year, so they join once names are reconciled. FAOSTAT is streamed from its 32 MB compressed archive and filtered to the seven stock series rather than loaded whole. Names are matched through `pycountry` with overrides for forms such as “Türkiye” and “Côte d'Ivoire”, and World Bank regional and global aggregates are excluded by requiring a three-character ISO code, so rows such as “World” never enter the panel as though they were countries.

The grid holds 6,700 country-year rows across 268 countries for 2000 to 2024. Three restrictions produce the modelling matrix: only equine-holding countries, defined as a maximum equid stock of at least one thousand; only 2005 to 2023; and only rows with an observable next-year label. The result has 3,078 rows, 44 features and 172 positives.

Events are de-duplicated on country, year and source record identifier, so one paper reporting several cases cannot contribute multiple counts. Missing values are median-imputed inside the training pipeline, with a missing-indicator feature per affected column, so the test period cannot influence imputed values. Missingness is low: 0.00% for FAOSTAT, 1.68% for the World Bank, 2.19% for climate, and 0.99% overall, with no column above 5.8%.

3. Label Definition

The target is binary: whether a glanders occurrence was documented in country i during year $t+1$. Occurrence is extracted from Europe PMC by matching, within one sentence, a country name from a curated gazetteer and a term from an occurrence lexicon (*outbreak*, *case*, *detected*, *seropositive*, *surveillance*). An event is recorded for the year in the sentence, or the publication year where none is stated. Because the label depends on publication it is termed *documented activity*: a negative may mean no disease or no report, so the model predicts surveillance-observable activity rather than biological presence.

4. Feature Engineering

Four groups are built from information available at year t or earlier. Seven disease-history features cover prior-year occurrence, a five-year total, years since the last event, and world and regional activity shares. Fourteen population features cover log equid and livestock stocks, equid share, density, growth and species counts. Ten climate features cover temperature, precipitation, humidity and wind, each with an anomaly against the country's own expanding mean, separating unusual conditions from latitude. Eleven socio-geographic features cover income, agriculture, GDP, rural share, connectivity and population, and largely proxy surveillance capacity.

Three derived features need definition. Equid share is horses, asses and mules divided by total farm animals. Livestock units apply weights of 0.8 for cattle and camels and 0.1 for small ruminants, so one number expresses grazing pressure across species. Years since the last event uses 99 to denote no event in the window.

Selection ran in two stages. Columns with mutual information below 0.002 and absolute Spearman correlation below 0.02 were removed, then one member of each pair correlated at or above 0.90, plus a column that was exactly the sum of three others, which a tree can construct itself. No projection was applied: with 44 features and 3,078 rows the binding constraint is signal rather than dimensionality, and Section 4. shows reduced sets transfer better.

5. Models

Six families are compared. Logistic regression is the additive baseline with balanced class weights. A single shallow decision tree is included because a non-specialist can read it. Random forest and extra trees are bagged ensembles [20], differing in split randomisation; XGBoost and LightGBM are boosted ensembles [18, 19]. Tuning used grouped cross-validation on country. Neural networks were excluded: with 172 positives and 44 features a network has more parameters than examples and would memorise the training period.

6. Evaluation Design

Three schemes answer different questions. A temporal holdout trains on 2005 to 2017 and tests on 2018 to 2023, with the test period never observed during fitting, which answers whether the model can be used in following years. Leave-country-out uses five folds grouped by country, so every prediction is for an unseen country; it answers whether the model transfers and it governs selection, being the harsher test and the closer approximation of deployment. Repeated stratified cross-validation, five folds three times, is a stability check.

Precision-recall AUC is the primary metric: at 5.59% prevalence ROC-AUC is dominated by the true-negative mass and overstates performance [22, 23], so it is reported only for comparability. Brier score and expected calibration error are reported because an uncalibrated probability cannot support a threshold-based alert [25, 26]. Precision, recall, specificity, F1 and the Matthews correlation use a threshold tuned on training data only, with bootstrap percentile intervals over one thousand resamples. Grouped rather than random splitting is essential, since a random row split would let the model memorise country identity [24].

7. Leakage Controls

Six controls apply: features use year t or earlier and only the target uses $t + 1$; the split is chronological; hyperparameters are tuned inside training folds; calibration is fitted on training folds only; the alert threshold is fixed on training; and imputation values and missing indicators are generated inside the pipeline.

Chapter 5

MLOps and Model Serving

1. From a Notebook to a Service

A trained model is not a component until it can be called, versioned, monitored and replaced. Three things must be true first. The model must be a portable artefact, so the object scored in evaluation is the object served, with nothing re-fitted in between. Every prediction must name the version that produced it. And the service must degrade predictably, so a bad release or a missing feature does not produce confident nonsense. The project is a sequence of numbered scripts, each producing artefacts consumed by the next, with every intermediate output cached so any stage can be re-run alone. That is also what makes the pipeline inspectable: a reader can run one stage and check it before trusting the next.

2. The Model Bundle

The unit of deployment is a directory, not a pickled object, because a pickle alone cannot say what its inputs mean.

```
gw-xgb-2026.11.1/
  model.joblib        # fitted pipeline, predict_proba
  manifest.json       # feature order, dtypes, calibration, metrics
  thresholds.json     # level boundaries and hysteresis margin
  SHA256SUMS
```

The manifest is what makes the bundle reusable. It records the ordered feature list with dtypes, so serving rejects a request with the wrong columns before any inference happens. It records the calibration method and achieved error, so a consumer knows whether to read the output as a probability or only as a rank. It records the evaluation metrics, so the number in the report and the number in the registry are the same. Because the bundle is a directory with a manifest, the API, a batch job and a notebook load it with identical behaviour.

Every training run also writes a record with the git commit, data version, feature list and hash, hyperparameters and the full metric set for all three schemes. Promotion is decided on leave-country-out precision-recall AUC rather than on the more flattering temporal holdout. A candidate becomes champion only if it beats the incumbent by more than the bootstrap interval width, its expected calibration error is below the deployment ceiling of 0.05, its feature schema matches what serving can supply when checked by name and order, and it has scored a shadow period without a prediction outside the plausible range. The previous champion is retained, so promotion is reversible, and a bundle failing its checksum or schema check is replaced by the previous champion rather than served.

3. Serving Design

The interface is deliberately coarse: a caller asks for a prediction for an area and period and receives the full record, rather than assembling one from several calls.

Table 5.1: HTTP interface of the risk engine.

Endpoint	Method	Purpose and contract
`/v1/predict`	POST	Score one area and period from a supplied or stored feature vector. Returns one `PredictionRecord`.
`/v1/predict/batch`	POST	Score up to 500 area-period pairs, with per-item errors rather than failing the request.
`/v1/risk-map`	GET	GeoJSON of the latest predictions for a period, filtered by risk level. Serves the dashboard directly.
`/v1/models`	GET	Registry listing: model id, promotion date, metrics, feature-schema hash.
`/v1/models/{id}`	GET	The manifest of one version, for inspection by a consumer.
`/v1/health`	GET	Liveness and readiness, where readiness checks that a champion bundle is loaded and its schema resolves.

Every response carries the `model_id` that produced it and a `data_freshness` block. The alert service reads that block before sending anything, so a prediction computed on stale upstream data cannot produce a message.

The order of operations is a reliability choice. Validation runs first, so a malformed request costs a schema check rather than a model load. The cache is consulted before inference because risk maps are read far more often than recomputed. Inference is deterministic for a fixed bundle, so a cache key of model id, area and period is safe, and the model id is part of the key rather than a version counter so promotion invalidates the cache. Responses are written back to the predictions table as well as cached, so the map does not depend on a warm cache.

The endpoint is specified against a budget rather than measured, since the service is not deployed: 40 ms on a cache miss and 4 ms on a hit, of which feature retrieval takes 25 ms and inference 8 ms. The explanation endpoint has a separate budget of two seconds because it includes generation, which is why the alert service never calls it inline. A message must not wait on an LLM.

4. Monitoring and Retraining

Three classes of signal are monitored. Service health covers request rate, error rate and latency percentiles per endpoint on a Prometheus endpoint charted in Grafana, which catches the service being down, the least interesting failure. Data drift compares each feature's current distribution with the training distribution using a population stability index and a Kolmogorov-Smirnov statistic, because a tree asked to extrapolate outside its training range returns a confident answer with no basis. Prediction drift and calibration decay track the mean predicted probability per period against the observed event rate as labels arrive: a model that keeps saying 0.2 while the observed rate falls to 0.05 has lost calibration even with its ranking intact.

Retraining follows from the third signal. Recalibrate when the observed-to-predicted ratio leaves a band for two consecutive periods, and retrain when a new year of labels is available and the champion's leave-country-out score has degraded beyond the bootstrap interval.

The pipeline runs linting, unit tests and a smoke test on every push. Unit tests cover what can be checked without data: schema validation, the level-threshold function including hysteresis at the boundary, idempotency key derivation, and the citation parser. A further test loads the committed bundle, scores a fixed fixture and asserts the output to four decimals, which catches the most common silent failure here, where a dependency upgrade changes numerical behaviour and every downstream number moves unnoticed.

The full computation runs in about 4.5 minutes on an Apple M4 processor with 10 cores and 16 GB of unified memory without a graphics accelerator. Per-family fitting time at the country-year level ranges from 1.9 seconds for the decision tree to 50.7 seconds for LightGBM. One module-level seed governs fitting, cross-validation, bootstrap resampling and SHAP sampling, with software versions Python 3.12.13, scikit-learn 1.9.1, XGBoost 3.4.1, LightGBM 4.7.0, SHAP 0.54.1, NumPy 2.5.4, pandas 3.1.1, SciPy 1.18.1, and the bundle smoke test runs single-threaded because tree ensembles are not bitwise reproducible across thread counts.

Chapter 6

The RAG Explanation Layer

1. The Problem This Layer Solves

A calibrated probability of 0.24 is actionable for a veterinary officer and meaningless to an equine owner. The owner needs to know what happened, what it means for their animals, and what to do next. The officer needs to know which factors drove the number, because a forecast they cannot interrogate is a forecast they will not act on.

Asking a large language model is the obvious answer and the obvious hazard. A model asked about glanders answers from training data that may be outdated, may conflate glanders with melioidosis, and may invent a treatment protocol. For a zoonosis under statutory control, an invented instruction is worse than no instruction. Retrieval-augmented generation [27] changes what the model may use: it is not asked what it knows, but given a small set of passages from official guidance plus the prediction's own numbers, and required to answer only from those. Each claim traces to a passage, and the passage is shown to the reader.

The goal is therefore narrow and testable: an answer a veterinary officer would accept as consistent with the National Action Plan, and a refusal rather than a guess when the corpus does not support one.

Fine-tuning would produce fluent text with no citations and no way to tell guidance from the base model's priors, and it would freeze the knowledge at training time, so a revised plan would require retraining. Retrieval keeps knowledge in a corpus that a re-ingestion can update, makes every answer auditable by construction, and allows a refusal path that fine-tuning does not. The cost is latency and possible retrieval failure, both handled explicitly below.

2. Knowledge Base

The corpus is restricted to sources with authority, because the layer's value is that it cites them. Each document carries an authority tier, used as a tie-breaker during retrieval.

Table 6.1: Knowledge base, by authority tier.

Source	Tier	Use
DAHD National Action Plan for Glanders, 2025 [34]	1	Case definitions, surveillance obligations, movement control, quarantine, notification duties.
WOAH and WHO disease information sheets	1	Disease biology, transmission, zoonotic risk, official status.
ICAR and national laboratory guidance	2	Diagnostic sampling, laboratory referral.
Open-access research papers	3	Epidemiology, risk factors, regional situation.

Tier 3 may inform an explanation but never overrides tier 1 on a control instruction. When passages conflict the higher tier wins and the conflict is logged, because a silent contradiction between sources is exactly the failure this layer exists to prevent.

3. Ingestion, Embedding and Retrieval

Ingestion runs offline as a job. A document is parsed, split at section boundaries, and each chunk is stored with enough metadata to be cited and filtered.

Chunking is section-aware rather than fixed-width, because a fixed window cuts control instructions in half and lets the model complete a truncated sentence. The splitter respects headings first, then paragraphs, and falls back to a sliding window only for a section longer than the ceiling. Chunks target 400 to 700 tokens with 15 per cent overlap inside a section, sections under 80 tokens merge into the parent, and the retained metadata is document id, title, full section path, page, year and authority tier. The section path is kept in full, for example `NAP-2025 > 5.2 Movement control`, so a citation points to a subsection rather than to a 90-page document.

Chunks are embedded with a compact sentence-transformer [28] producing 384-dimensional vectors, cheap enough to embed on a laptop, with a multilingual variant for Hindi because an English-only embedding degrades on Devanagari. Vectors live in pgvector beside the relational tables, so re-ingestion and prediction share one transaction boundary. The index is HNSW with cosine distance [29], acceptable because the reranker corrects recall at the head of the list and it keeps query cost flat as the corpus grows.

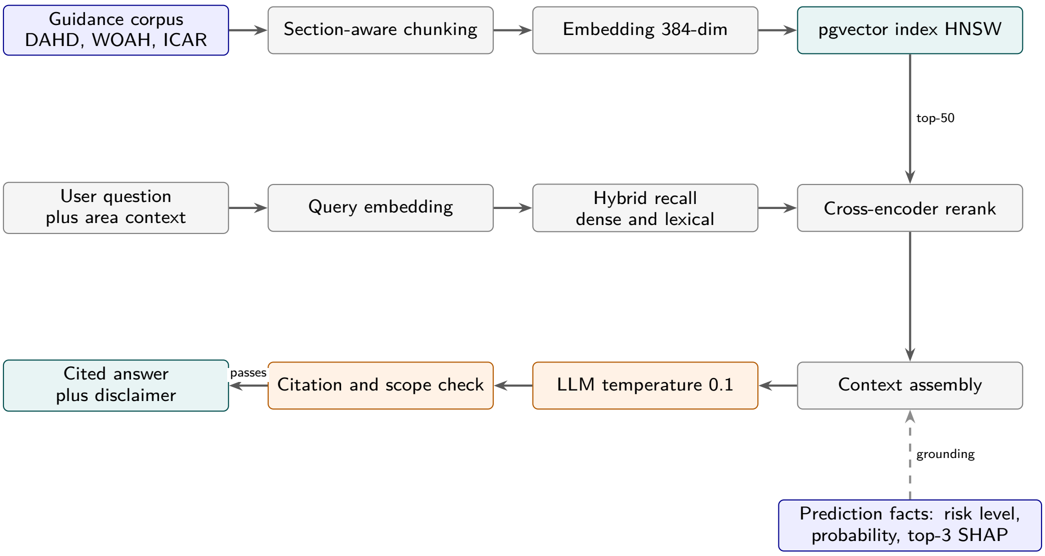
Guidance corpus
DAHD, WOAH, ICAR
Section-aware chunking
Embedding 384-dim
pgvector index HNSW
top-50
User question
plus area context
Query embedding
Hybrid recall
dense and lexical
Cross-encoder rerank
Cited answer
plus disclaimer
passes
Citation and scope check
LLM temperature 0.1
Context assembly
grounding
Prediction facts: risk level,
probability, top-3 SHAP

Figure 6.1: The RAG layer. Ingestion (top row) is an offline job; retrieval, assembly and generation (below) run per request. Predicted risk and SHAP factors are injected as structured facts, so the answer explains this forecast rather than glanders in general. The citation check sits after generation and can return a refusal instead of an unsupported answer.

Retrieval is hybrid, because dense and lexical search [30] fail differently here. A dense query for “what should I do if I suspect glanders” retrieves passages about suspicion, which helps. A dense query for “movement control” may miss a passage using “restriction of equid movement”, where a lexical match succeeds; conversely a lexical query for a rare acronym retrieves noise a dense query ignores. Both lists are retrieved and fused by reciprocal rank fusion [31], which combines rankings without requiring comparable scores:

$$\mathrm{RRF}(d) = \sum_{i \in \{\text{dense},\text{lexical}\}} \frac{1}{k + \mathrm{rank}_i(d)}, \qquad k = 60$$

The fused top fifty are reranked by a cross-encoder scoring each passage against the question jointly. A cross-encoder is far slower per pair than an embedding dot product, which is why it runs on fifty candidates rather than on the corpus, and the top five become the context.

The relevance floor drives the refusal path: if fewer than two passages clear it, the layer returns a refusal with the SHAP factors, because a context built from marginal passages produces a confident answer built on nothing.

Table 6.2: Retrieval parameters.

Stage	**Setting**	**Reason**
Dense recall	top 30, cosine	Semantic paraphrase coverage
Lexical recall	top 30, BM25	Exact terms, acronyms, section titles
Fusion	reciprocal rank, $k = 60$	Scale-free combination of two rankings
Rerank	cross-encoder, top 50 to 5	Precision at the head of the list
Relevance floor	rerank score below 0.30 discarded	Prevents padding the context with weak passages

4. Context, Generation and Guardrails

The prompt has three parts in fixed order: a system instruction, the retrieved passages with identifiers, and the structured prediction facts. Those facts make the answer specific to this forecast rather than to glanders in general, and they are injected as structured data so the model cannot restate an association as a cause. The system instruction states four constraints: answer only from the passages, cite every claim, never diagnose an individual animal, and state that the output is a population-level forecast.

Generation uses temperature 0.1 and a bounded output length. Low temperature is chosen for consistency rather than caution: two officers asking the same question about the same area should receive the same instruction. The provider sits behind a single-method interface, so the same retrieval, grounding and citation code runs against a local Ollama model or a hosted API, and switching model cannot silently disable the guardrails.

The layer is designed to refuse, and four checks can terminate a response.

Table 6.3: Guardrails and their failure behaviour.

Check	**What it verifies**	**On failure**
Retrieval floor	At least two passages above the relevance threshold	Refusal with the top SHAP factors, which need no retrieval
Citation resolution	Every cited identifier exists in the retrieved set	Answer rejected and regenerated once, then refusal
Diagnosis scope	No individual-animal diagnosis or treatment prescription	Response rewritten without the clinical claim, and the attempt logged
Out-of-scope	Question is about glanders control, not unrelated topics	Refusal stating what the layer covers

The citation check is mechanical. The answer is parsed for identifiers and each is looked up in the retrieved set; one that does not resolve means the model cited a document it was not given, which is a hallucination whether or not the sentence is true. Asking a model to be accurate is not a control; checking its output is.

5. Evaluation

Four quantities are measured, none of them whether the prose reads well. Retrieval quality is recall@5 and mean reciprocal rank on a hand-built question set, which isolates retrieval from generation, since no generation quality helps if the right passage is never retrieved. Faithfulness is whether each claim is supported by the retrieved context, scored by a judge model and reviewed by hand on a sample; this catches a model that cites a passage but says something it does not say. Citation precision is the fraction of citations resolving to a passage that supports the sentence, which is distinct from resolution, since resolution only checks existence. Refusal behaviour is the refusal rate on out-of-scope questions and the false refusal rate in scope: a layer that never refuses is unsafe and one that always refuses is useless, so both are reported.

Because the layer is specified rather than deployed, the validation that would precede deployment is stated so the claim is falsifiable: recall@5 above 0.85, faithfulness above 0.90 on 100 manually reviewed answers, citation precision above 0.95, zero individual-animal diagnoses on an adversarial probe set, refusal on every out-of-scope question with a false refusal rate below 10 per cent, and p95 latency under two seconds. These are targets, not results.

Chapter 7

The Alert Service

1. Why the Alert Is the Hard Part

The risk engine produces a probability and the RAG layer explains it, but neither reaches the person who owns the animals, whose only interaction is a message on a phone competing with everything else on it. The design problem is therefore not how to send a message but how to send few enough that the ones which arrive are read. A system that warns about every area every month trains its users to ignore it, then fails silently at the moment it was built for.

The service is event-driven: the prediction job publishes `predictions_ready` and the evaluator consumes it, so an alert cannot be sent for a period whose predictions were not recomputed. For each subscribed area the evaluator reads the new prediction and the previous alert state, and those two values are the entire input to the rules. It reads no features or model internals, so a model change can affect alert behaviour only through the probability it produces.

2. Rule 1: Threshold with Hysteresis

The calibrated probability maps to three levels. The boundaries are configuration values derived from the operating curve in Chapter 8, not defaults and not 0.5.

Table 7.1: Level thresholds and their basis.

Level	**Band**	**Meaning for the recipient**	**Basis**
LOW	below the medium boundary	No message. Routine monitoring.	Below the budget where precision collapses
MEDIUM	medium to high boundary	Prepare; check animals; note the veterinary contact	Operating point where recall exceeds 80 per cent
HIGH	above the high boundary	Act: restrict movement, isolate suspects, inform the authority	Top of the operating curve, precision above 0.5

Hysteresis addresses a specific defect. With a single boundary, an area whose probability oscillates around the threshold produces an alert every period, because each crossing is a state change. Two boundaries are therefore used: an area enters HIGH at the high threshold and leaves only when the probability falls below that threshold minus a margin.

$$\text{enter HIGH at } p \geq T_H, \qquad \text{leave HIGH only at } p < T_H - \delta, \qquad \delta = 0.05$$

The margin is configurable per disease. A larger margin gives a more stable state at the cost of slower de-escalation, which suits a disease where a missed warning costs more than a late all-clear.

3. Rule 2: Suppression

Suppression decides whether a message is warranted at all, given that the level has changed. Six conditions apply and any one can stop the message.

Table 7.2: Suppression conditions.

Condition	Rule	Rationale
State change	Send only on a transition, not while the level is held	A monthly reminder at the same level carries no new information
Cooldown	No repeat of the same level within a configurable window	Prevents oscillation at a boundary from producing repeated messages
Escalation bypass	An upward transition ignores the cooldown	A worsening must never be delayed by a timer
Per-user daily cap	At most one message per recipient per day	Protects against several areas escalating at once
Quiet hours	Nothing sent between 21:00 and 07:00 local; queued instead	A warning that wakes a household at night is resented, and is not actionable then
Stale-data guard	No send if `data_freshness.passes` is false	An alert about a prediction computed on stale inputs is worse than no alert

A manual mute per area lets an officer silence a zone under active surveillance without disabling the service. De-escalation is separate: when a level falls, an optional short all-clear is sent, since a recipient told to act needs to be told when to stop. Sending it is a configuration choice, because an all-clear costs a message but prevents a restriction being maintained indefinitely.

4. Rule 3: Deduplication

The service must be safe to re-run: jobs are retried, deployments restart workers, and a model may be re-scored for the same period, and none may produce a second message. Deduplication is enforced at three levels, each with its own key and database constraint.

Table 7.3: Deduplication levels.

Level	Key	What it prevents
Alert	`(area_id, period, level)`	A re-run of the job creating a second alert for the same period
Recipient	`(alert_id, subscription_id)`	One subscriber receiving the same alert through two fan-out passes
Send	`(alert_id, subscription_id, attempt)`	A retry after a timeout delivering a message that had in fact succeeded

The first two are unique database constraints, so protection does not depend on application logic being correct. The alert identifier is derived deterministically from its key rather than at random, so two concurrent workers computing the same alert arrive at the same identifier and one loses the insert. The send-level check reads the delivery log before each attempt, handling the case where a provider accepts a message, the response times out, and the retry would deliver it twice.

5. Fan-Out, Channels and Delivery

Once an alert record exists, one delivery job per subscription is queued through Redis. A channel adapter receives a rendered message and a recipient and returns a delivery status, with adapters for SMS, WhatsApp and Telegram, the last used for demonstration because it needs no commercial onboarding. Retries use exponential backoff with jitter: a permanent failure such as an invalid number is not retried, a transient one such as a provider timeout is, and the distinction is logged, because a silent permanent failure leaves a subscriber who believes they are covered and is not. Templates are separate from the rules engine, so adding a language adds a template set rather than touching the rules.

The same alert through a richer channel carries the explanation paragraph from the RAG layer with its citations. The rule that decides whether to send is identical; only the adapter and template differ, which is what lets one alert serve a feature phone and a smartphone without duplicating the decision logic.

6. Evaluation of the Alert Layer

The service is evaluated as a warning system, not as a messaging system. Alert precision is the fraction of HIGH area-periods followed by a documented event, and it determines whether recipients trust the message. Event coverage at budget is the operating curve already reported in Chapter 8, and the thresholds are read from it. Noise per recipient is messages per subscriber per year, which predicts whether the service will still be read in its second year. Delivery reliability is the fraction of queued deliveries reaching a terminal success state within the retry window, reported per channel. The first two come from historical predictions and labels; the last two need a deployed service and are stated as the protocol that would measure them.

Chapter 8

Results and Discussion

1. Dataset Characteristics

The modelling matrix holds 3,078 country-year observations, of which 172 are positive: a prevalence of 5.59% and an imbalance of 16.9 to one. 56 of 162 countries record at least one positive, 30 of them in two or more years, and the top five hold 35.5% of all positives. The imbalance rules out accuracy as a metric and motivates precision-recall analysis; the persistence justifies historical features; and the concentration is the first indication that the label tracks reporting capacity, since those countries operate reference laboratories.

2. Model Comparison

Table 8.1 reports discrimination under all three schemes.

Table 8.1: Discrimination at the country-year level. Brackets give 95 per cent bootstrap intervals on the temporal holdout.

Model	PR-AUC (time)	95% CI on PR-AUC	ROC-AUC (time)	PR-AUC (country)	PR-AUC (rep. CV)	Brier	Prec.	Rec.
Logistic regression	0.454	[0.34, 0.58]	0.812	0.223	0.414	0.0547	0.25 / 0.57	
Decision tree	0.389	[0.29, 0.50]	0.861	0.267	0.360	0.0602	0.49 / 0.53	
Random forest	0.554	[0.43, 0.66]	0.870	0.304	0.476	0.0503	0.53 / 0.50	
Extra trees	0.530	[0.42, 0.64]	0.854	0.363	0.476	0.0535	0.47 / 0.51	
XGBoost	0.485	[0.37, 0.60]	0.844	0.281	0.423	0.0536	0.61 / 0.38	
LightGBM	0.496	[0.38, 0.61]	0.841	0.339	0.397	0.0534	0.65 / 0.30	

Against a base rate of 5.59%, the strongest temporal result of 0.554 is an approximately 10-fold lift. Every tree-based model beats the logistic baseline, and the single decision tree is weakest, which indicates that the relationships are interactive rather than additive. The more important pattern is the decline under leave-country-out validation: for Random Forest the score falls from 0.554 to 0.304, and the decline exceeds the bootstrap interval width for every family. That is the signature of learning country-specific structure, and the estimate to quote for deployment.

3. Calibration

Predictions are well calibrated, with an expected calibration error of 0.016 and a Brier score of 0.0535: a stated risk of twenty-four per cent corresponds to an observed frequency near twenty-four per cent. That is what makes a threshold-based alert defensible, since a message quoting a probability must be numerically trustworthy.

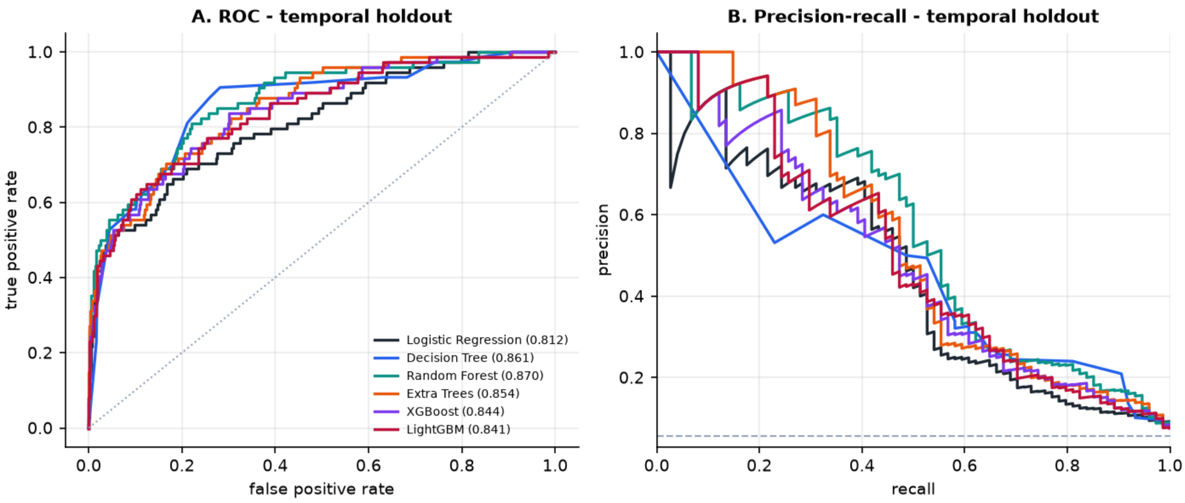

Figure 8.1: ROC curves (panel A) and precision-recall curves (panel B) on the temporal holdout. At this prevalence the ROC panels compress into a narrow band while the precision-recall panels separate the families.

4. Ablation

Feature groups were removed and each configuration retrained (Table 8.2).

Table 8.2: Ablation settings, using XGBoost with fixed hyperparameters.

Feature set	**Columns**	**PR-AUC (time)**	**PR-AUC (country)**
History only	7	0.407	0.174
Population only	14	0.432	0.241
Climate only	10	0.395	0.078
Socio-geography only	11	0.469	0.160
History and population	21	0.451	0.321
All groups	42	0.487	0.259
All groups plus surveillance proxy	44	0.497	0.261

Two results matter more than the headline. The reduced set of 21 columns reaches 0.321 under leave-country-out validation, against 0.259 for the full 42: added covariates improve in-distribution fit and reduce transfer by letting the model fit country peculiarities. Climate alone reaches 0.078, marginally above the 5.59% floor, so annual country climatology largely encodes latitude and therefore country identity. Adding research-output proxies moves the leave-country-out score by +0.001, weak evidence against pure publication tracking but not proof.

5. Chained Pipeline

Table 8.3 reports the effect of appending the stage-one population forecast.

Table 8.3: Effect of chaining the population forecast into the occurrence model.

Variant	**PR-AUC (time)**	**ROC-AUC (time)**	**PR-AUC (country)**
Plain (no chained feature)	0.530	0.854	0.363
Chained (population forecast added)	0.527	0.853	0.344

The population forecast performs well as a forecast, with a mean absolute error of 0.424 on the log scale and a Spearman correlation of 0.976. The intensity stage is weaker but usable, at a mean absolute error of 0.150 and a rank correlation of 0.465 among positive observations. The hurdle product ranks expected annual burden with a Spearman correlation of 0.326, and its highest decile captures 70 of 106 observed events.

Appending the forecast changes temporal PR-AUC by -0.003, inside the noise band. The cause is structural: population magnitude is already a direct feature, so forecasting it supplies a less accurate copy of what the model holds. Chaining should help where that quantity cannot be observed directly, which is not the case here.

6. Alert Economics

As a warning system, alerting the six per cent of country-years at highest predicted risk captures about half of all observed events, and twenty-eight per cent captures eighty per cent (Figure 8.2). Uniform alerting would need full coverage for the same recall, so the model is roughly eight times more efficient than arbitrary selection at the fifty per cent point.

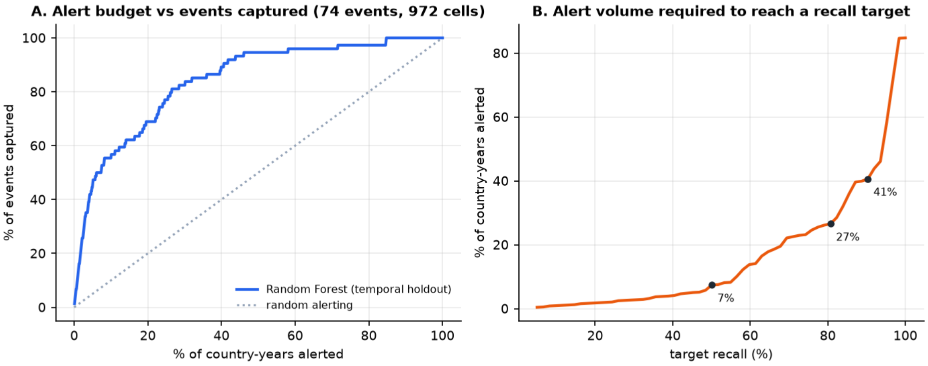

Figure 8.2: Alert economics. Panel A gives the fraction of events captured as the alert budget grows, against random alerting. Panel B gives the alert volume required to reach a target recall.

7. Subgroup Analysis

Table 8.4 reports leave-country-out performance by region. Skill is lowest in Sub-Saharan Africa and in low-income countries, where early warning would help most. The cause is label sparsity rather than a model defect: few events are recorded in those strata to learn from.

Table 8.4: Leave-country-out performance by World Bank region.

Region	Observations	Events	ROC-AUC	PR-AUC
East Asia & Pacific	456	48	0.865	0.396
Europe & Central Asia	912	31	0.826	0.163
Latin America & Caribbean	551	12	0.903	0.381
MENA, Afghanistan and Pakistan	380	23	0.834	0.220
Sub-Saharan Africa	665	27	0.699	0.073

8. Discussion

The model has predictive skill: a precision-recall AUC of 0.554 against a base rate of 5.59% is not explainable by chance, survives a forward-looking split, and is calibrated to within two points. It does not transfer intact, though: leave-country-out performance is roughly forty per cent lower, and that is the planning figure, so a new country needs local validation first.

The principal signal is surveillance rather than biology, and three analyses agree: the highest features are population and connectivity measures, the label concentrates in countries with reference laboratories, and the best-scoring cells are India, the United States and Australia. Climate, treated as a primary input in the proposal, transfers poorly here. Simpler models transfer better too, which argues for parsimony when positives are few.

Chapter 9

Conclusion and Future Work

1. Conclusion

This project set out to build a usable warning system for glanders: to establish whether population-level risk could be predicted from public data, and to design how a prediction reaches the person who owns the animals. Four sources not previously applied to the disease formed a country-year panel across 162 equine-holding countries, six model families were trained under three validation schemes, and a three-stage chained pipeline was tested. Around the model, delivery was decomposed into three components with fixed contracts: a risk engine, a retrieval-augmented explanation layer and an alert service.

Prediction is possible, and the performance is moderate. On a forward-looking temporal holdout the strongest model reaches a precision-recall AUC of 0.554 against a base rate of 5.59%, an approximately 10-fold lift, with a calibration error of 0.016, capturing half of all observed events while alerting six per cent of country-years. The skill is fragile across geography, though: leave-country-out performance is about forty per cent lower, and the decline concentrates where early warning would help most.

What the model learns is less encouraging. Three analyses converge: the highest-ranked features are population size and reporting capacity rather than biological drivers, the label concentrates in countries operating reference laboratories, and the best-scoring cells have the strongest publishing record. It predicts where glanders will be documented, not where the organism is present. The chained design does not improve discrimination because the intermediate quantity is already a direct feature.

The decomposition also proved useful: three components with versioned contracts, none hardcoding the disease, were each testable alone, so the risk engine could be evaluated before any explanation existed and the alert rules checked against a policy before any message was sent.

2. Future Work

Reported activity. The most important extension is to model surveillance capacity as a quantity, as an offset on publication volume or through a two-part model estimating detection and occurrence jointly; otherwise gains in score reflect reporting rather than disease.

The label extractor. The 263 extracted events have not been checked against a hand-annotated set; annotating a stratified sample would give the extractor's own precision and recall, bounding label noise that propagates downstream.

Resolution. The proposal envisaged district-month resolution for India; public district covariates would need assembling, but that resolution matches the scale at which control is applied.

The unmeasured components. The calibration here is one point estimate; production needs drift monitoring and a retraining trigger. The RAG layer's retrieval, guardrail and protocol targets are specified but unmeasured, and running it against the corpus would give recall@5, faithfulness, citation precision and the false refusal rate. **The field.** Messages per subscriber per year decides whether the service survives a second year; a district pilot would supply it. A WOAH WAHIS record set [33] would validate the literature-derived label where the two overlap, and a cost-sensitive threshold would match the operating point to the programme's priorities.

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